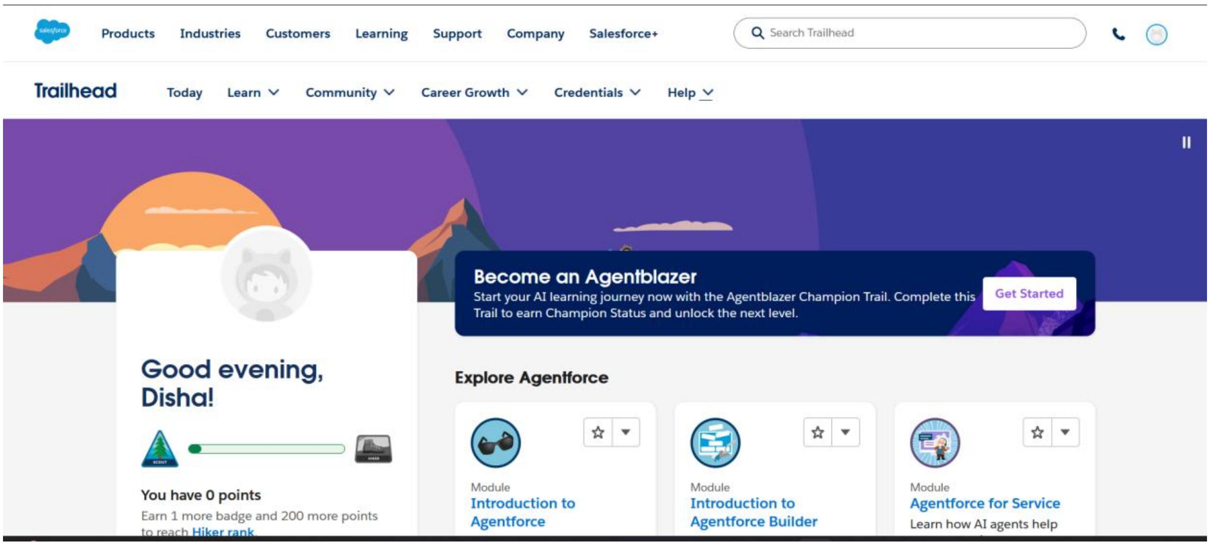
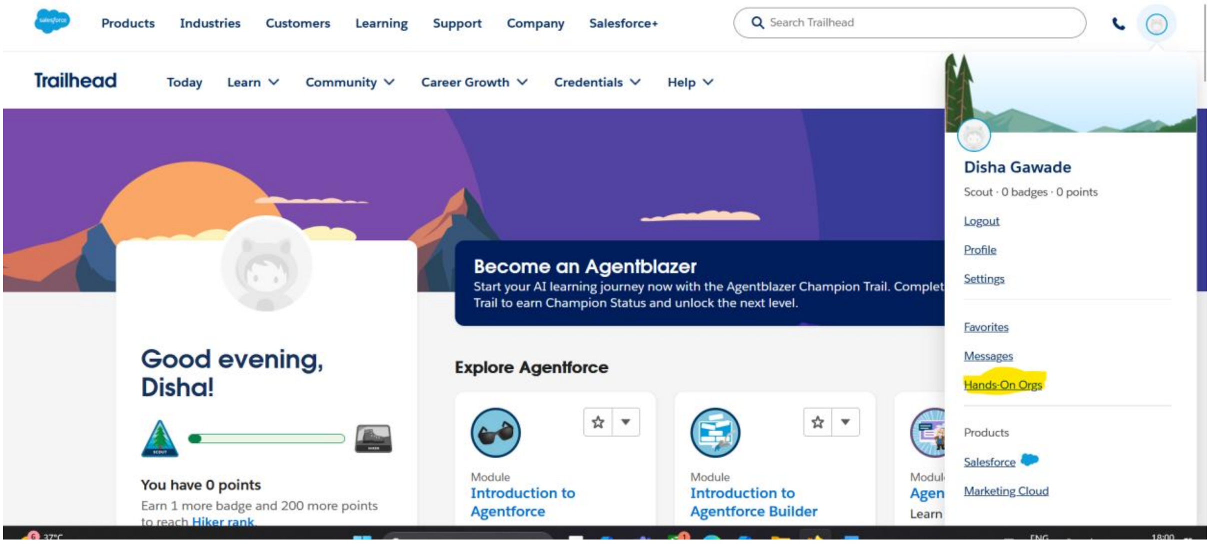
19.Creating an application in salesForce.com using Apex Programming Language

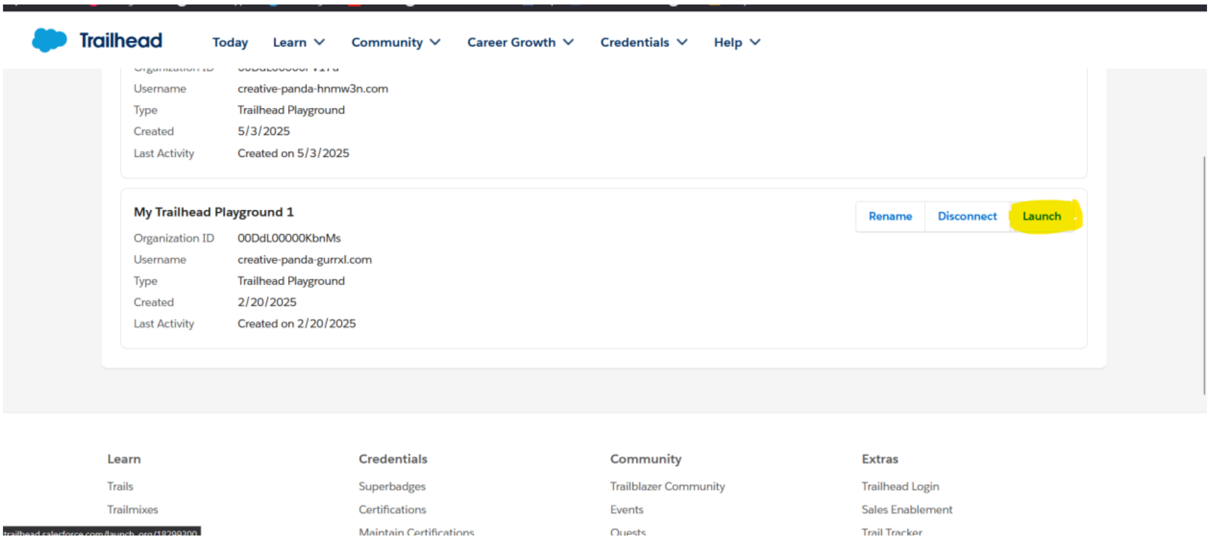
.Print message “Welcome to Apex Programming Language of sale force Platform”.

Step1 :- Create account and login

https://trailhead.salesforce.com/today






Trailhead
Today
Learn
Community
Career Growth
Credentials
Help
Username
creative-panda-hnmw3n.com
Type
Trailhead Playground
Created
5/3/2025
Last Activity
Created on 5/3/2025
My Trailhead Playground 1
Rename
Disconnect
Launch
Organization ID
00DdL00000KbnMs
Username
creative-panda-gurrxl.com
Type
Trailhead Playground
Created
2/20/2025
Last Activity
Created on 2/20/2025
Learn
Trails
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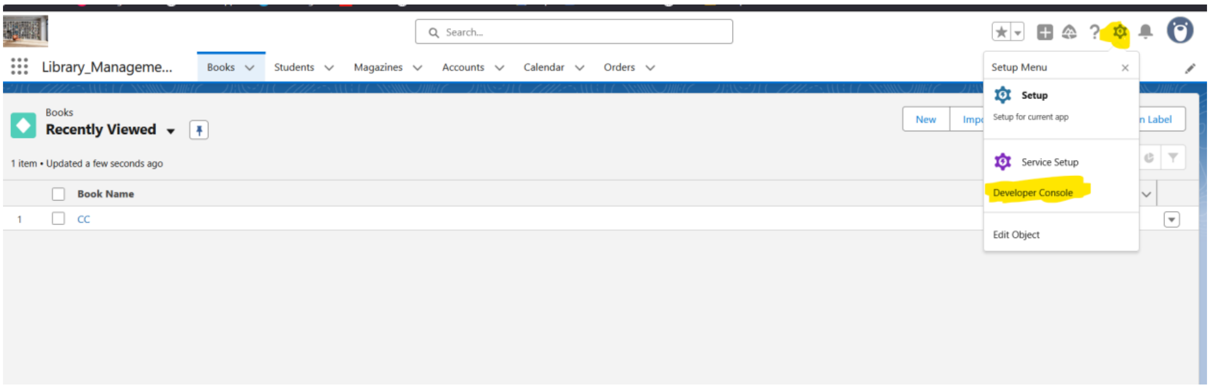
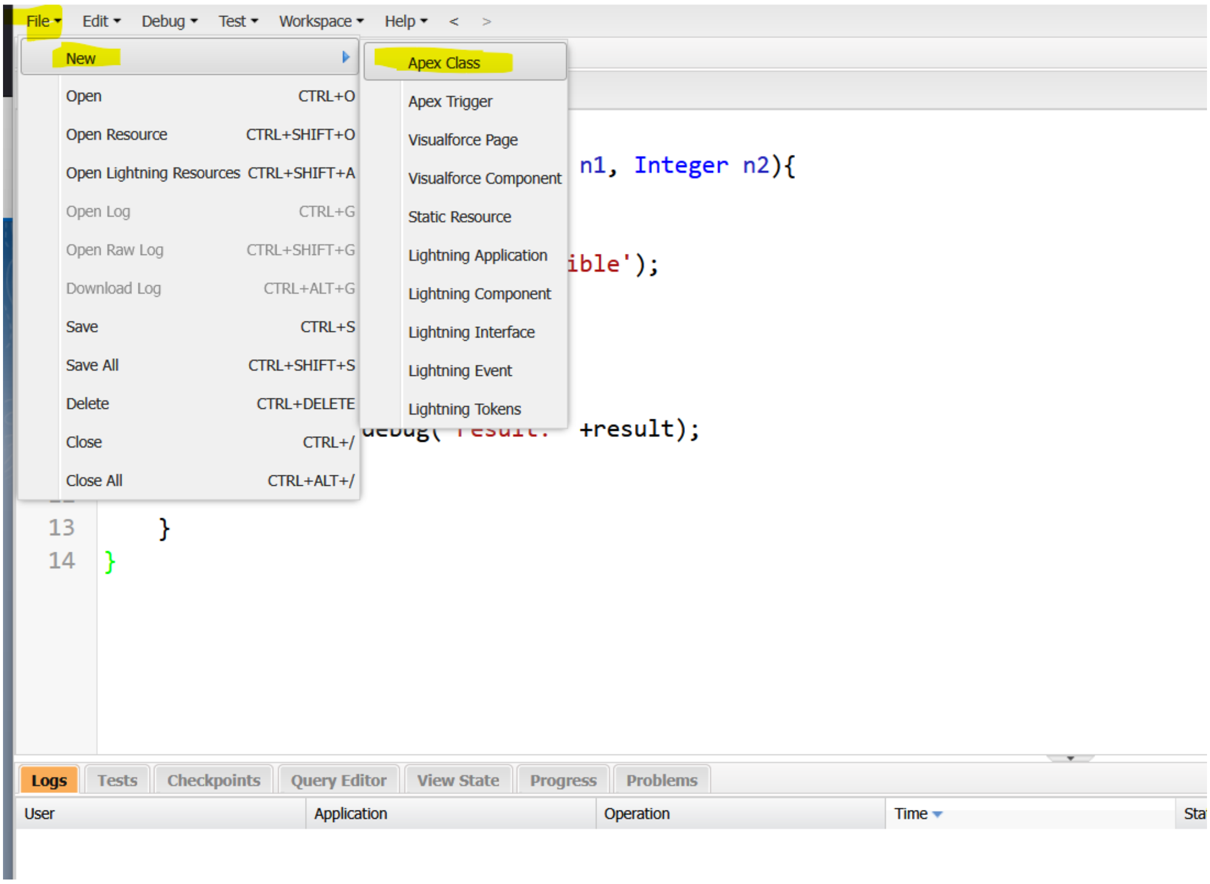
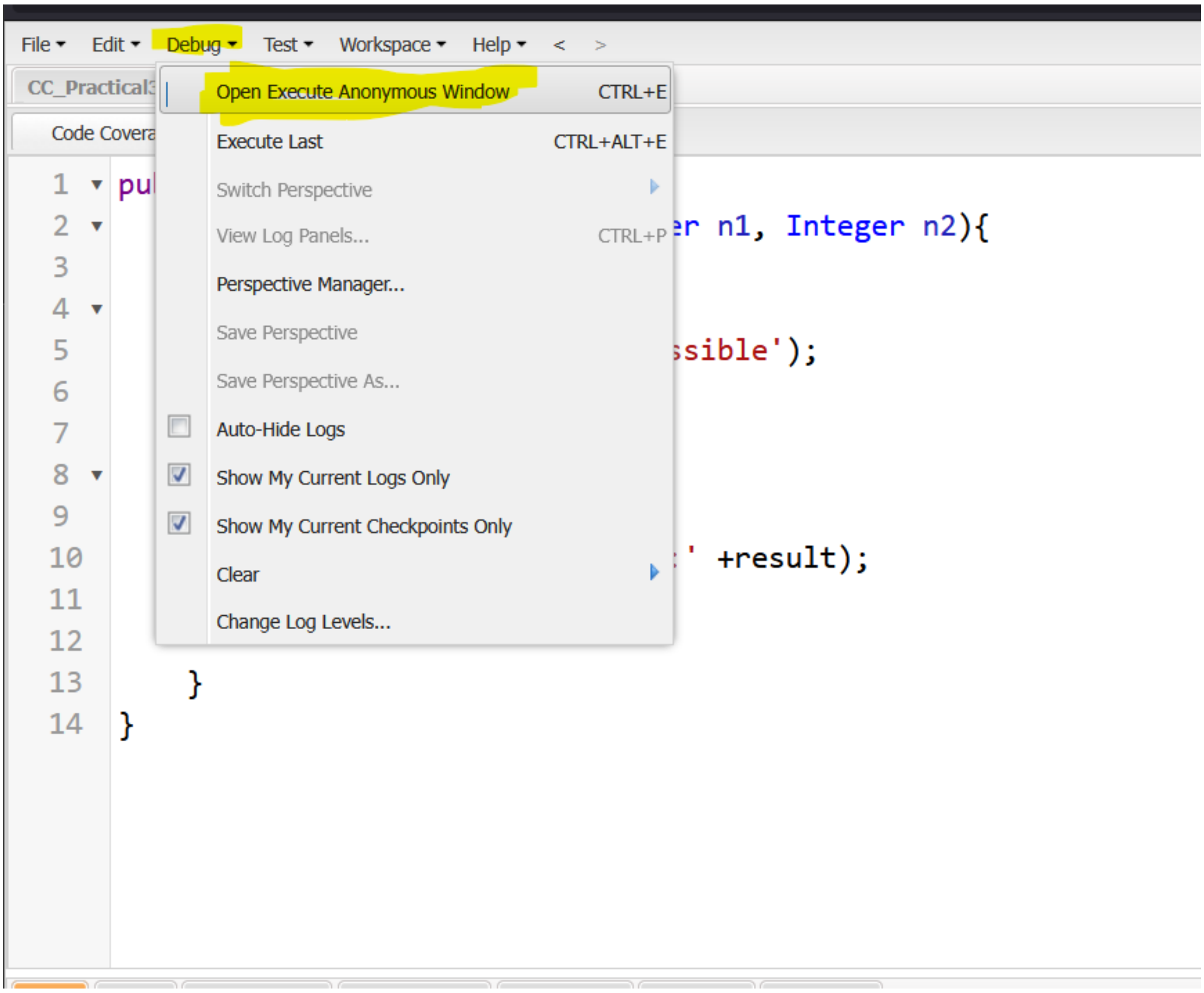
Credentials
Superbadges
Certifications
Community
Trailblazer Community
Events
Extras
Trailhead Login
Sales Enablement

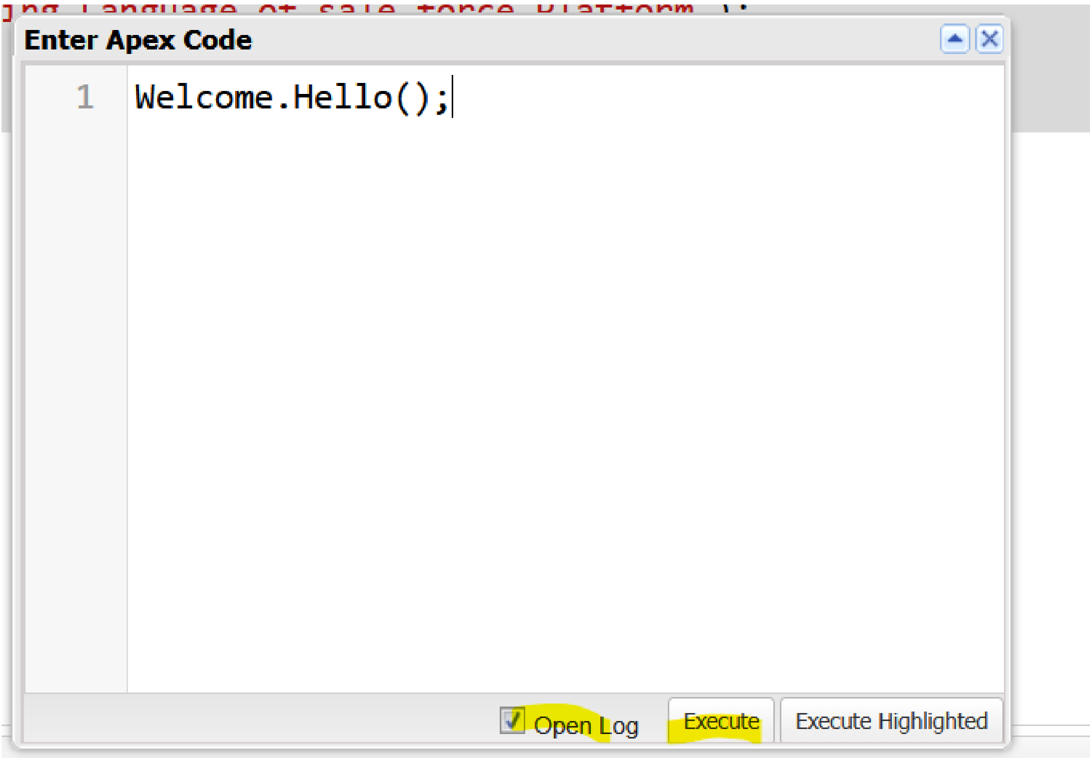

Library_Manageme...
Books
Students
Magazines
Accounts
Calendar
Orders
Books
Recently Viewed
1 item • Updated a few seconds ago
Book Name
cc
New
Setup Menu
Setup
Setup for current app
Service Setup
Developer Console
Edit Object

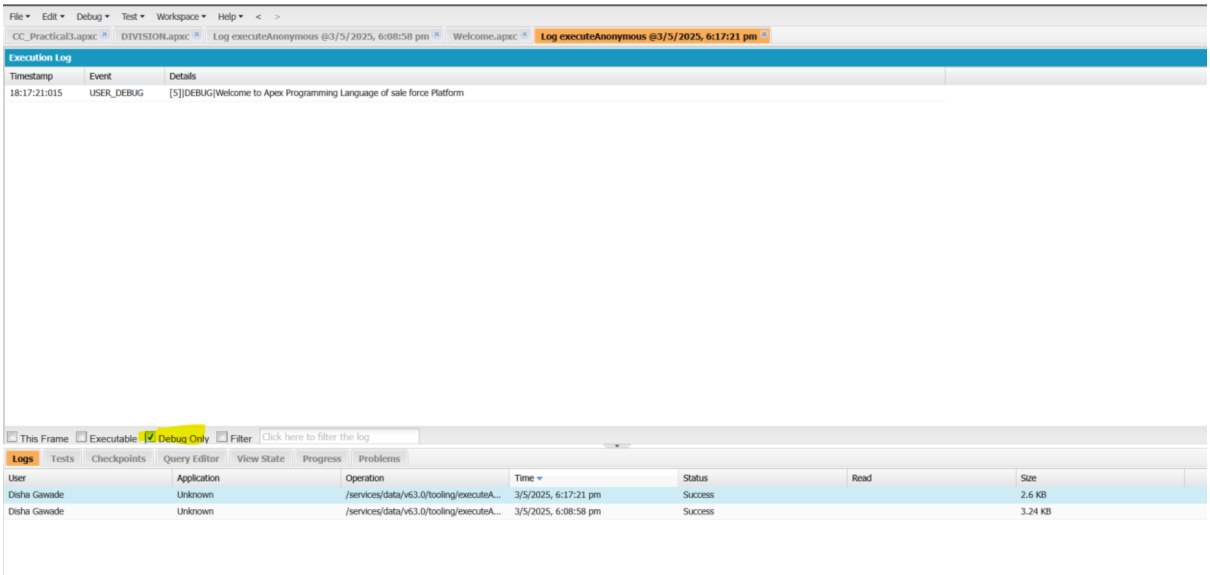

File
Edit
Debug
Test
Workspace
Help
New
Open CTRL+O
Open Resource CTRL+SHIFT+O
Open Lightning Resources CTRL+SHIFT+A
Open Log CTRL+G
Open Raw Log CTRL+SHIFT+G
Download Log CTRL+ALT+G
Save CTRL+S
Save All CTRL+SHIFT+S
Delete CTRL+DELETE
Close CTRL+/
Close All CTRL+ALT+/
Apex Class
Apex Trigger
Visualforce Page
Visualforce Component
Static Resource
Lightning Application
Lightning Component
Lightning Interface
Lightning Event
Lightning Tokens
n1, Integer n2){
ible');
+result);
}
}
Logs
Tests
Checkpoints
Query Editor
View State
Progress
Problems
User
Application
Operation
Time

Create apex class and write this code

```
public class Welcome {
  public static void Hello()
  {
    System.debug('Welcome to Apex Programming Language of sale force Platform');
  }
}
```







Q1.Detail introduction about salesforce trailhead.

Salesforce Trailhead is an innovative, interactive learning platform designed by Salesforce to help individuals and organizations learn about Salesforce technologies, tools, and methodologies. It serves as an accessible, gamified, and community-driven environment that caters to a wide range of users, from absolute beginners to seasoned professionals.

Key Features of Salesforce Trailhead

1. **Learning Paths (Trails)**

    - Trailhead offers structured learning paths called **Trails**, which guide users through topics like Salesforce administration, development, architecture, and more.
    - Trails are categorized based on skill level (beginner, intermediate, advanced) and roles (e.g., Admin, Developer, Marketer).
2. **Modules**
    - Each module in Trailhead focuses on a specific topic, such as CRM basics, Apex programming, or Einstein Analytics.
    - Modules are interactive and include bite-sized content, quizzes, and hands-on challenges.
3. **Hands-On Challenges**
    - Users apply what they've learned in a real Salesforce environment using a free Developer Org or Trailhead Playground.
    - These challenges assess knowledge retention and provide practical experience.
4. **Superbadges and Certifications**
    - **Superbadges** are advanced projects that simulate real-world scenarios, helping users showcase their expertise.
    - Trailhead also aligns closely with Salesforce certification programs, helping learners prepare for exams like Salesforce Administrator or Platform Developer.
5. **Gamification**
    - Users earn points and badges as they complete modules, trails, and challenges.
    - This gamified approach motivates learners and tracks progress.
6. **Career Opportunities**
    - Trailhead provides resources like **Trailblazer Community** and **Salesforce Talent Alliance** to connect learners with job opportunities.
    - The **Trailblazer Career Path** offers guidance for transitioning into Salesforce careers, with job-specific trails and networking events.
7. **Accessibility and Inclusivity**
    - Trailhead is free, making it accessible to anyone with an internet connection.

    - The platform emphasizes diversity and inclusion, catering to individuals from all backgrounds and skill levels.

Who Can Benefit from Trailhead?

1. **Beginners:**
    - Those new to Salesforce or the tech industry can start with foundational modules to build their knowledge.
2. **Salesforce Professionals:**
    - Administrators, developers, architects, and marketers use Trailhead to stay up-to-date with Salesforce innovations.
3. **Business Users:**
    - Non-technical users, like sales and service representatives, can learn how to effectively use Salesforce tools.
4. **Organizations:**
    - Companies can upskill employees using Trailhead for Teams, enabling them to maximize their Salesforce investment.

Why Use Trailhead?

- **Free and Flexible:** No subscription fees; learn at your own pace.
- **Practical Learning:** Gain real-world skills by working on hands-on challenges.
- **Certification Prep:** Aligns with Salesforce certifications, boosting career credentials.
- **Community Support:** The Trailblazer Community offers networking, mentorship, and collaboration opportunities.

Getting Started

1. **Sign Up:** Visit Trailhead's website and create a free account.
2. **Pick a Trail:** Start with trails like "Salesforce Basics" or "Admin Beginner."
3. **Earn Badges:** Complete modules, challenges, and projects to earn recognition.
4. **Join the Community:** Connect with other learners and Salesforce professionals through Trailblazer Community groups.

Trailhead is not just a learning platform—it's a career accelerator that empowers individuals to thrive in the Salesforce ecosystem while contributing to the vibrant Trailblazer Community.

Q2.introduction about Apex Programming Language

Introduction to Apex Programming Language

Apex is a strongly-typed, object-oriented programming language developed by Salesforce for building complex business logic and customizing applications on the Salesforce platform. It is deeply integrated into the Salesforce environment, allowing developers to write and execute code directly on the Salesforce platform.

Apex is designed to interact with Salesforce's built-in features, such as the data model, workflow automation, and security, making it a powerful tool for extending Salesforce functionalities.

Key Features of Apex

1. **Tight Integration with Salesforce**
    - Apex is natively integrated with Salesforce's multi-tenant architecture.
    - It allows direct access to Salesforce objects, records, and metadata using standard APIs and queries (SOQL and SOSL).
2. **Object-Oriented Programming**
    - Apex supports core OOP principles like classes, inheritance, interfaces, and polymorphism.
    - It provides a structured approach to coding that ensures code reusability and maintainability.
3. **Database Integration**
    - Apex has built-in support for database operations (DML: Data Manipulation Language).
    - It includes transactional control, rollback mechanisms, and the ability to handle bulk operations efficiently.
4. **Event-Driven Architecture**
    - Apex triggers respond to events such as data changes (insert, update, delete) on Salesforce objects.
    - This enables real-time automation and processing.

5. **SOQL and SOSL**
    - Apex includes Salesforce Object Query Language (SOQL) and Salesforce Object Search Language (SOSL) for querying and searching Salesforce records.
6. **Multitenancy Support**
    - Apex is designed to work within Salesforce's multi-tenant environment, enforcing governor limits to ensure fair resource usage.
    - These limits include constraints on CPU time, memory, database operations, and API calls.
7. **Built-in Testing Framework**
    - Apex has a native testing framework that allows developers to write test classes and methods.
    - At least 75% of Apex code must be covered by tests before deploying to production.
8. **Dynamic Typing and Scripting**
    - Supports dynamic SOQL, SOSL, and DML queries, offering flexibility for developers to construct queries at runtime.
9. **Scalability and Performance**
    - Apex is optimized for handling large data volumes and complex logic while adhering to Salesforce's best practices.

Use Cases of Apex

1. **Custom Business Logic**
    - Automate processes such as updating related records, calculating discounts, or sending notifications.
2. **Triggers**
    - Build before or after triggers to execute specific actions when data is inserted, updated, or deleted.
3. **Batch and Scheduled Jobs**
    - Perform long-running or large-scale operations using batch Apex.
    - Schedule jobs to run at predefined intervals using scheduled Apex.
4. **Web Services and APIs**

    - Expose Apex classes as REST or SOAP web services for external integrations.
    - Consume external services directly within Salesforce.
5. **Custom User Interfaces**
    - Use Apex in conjunction with Visualforce or Lightning Components to create tailored UI experiences.
6. **Error Handling and Validation**
    - Implement robust error handling and enforce complex validation rules.

Advantages of Apex

1. **Platform Integration:** Direct access to Salesforce’s CRM features and data.
2. **Productivity:** Simplifies development with features like pre-built libraries and Salesforce-specific methods.
3. **Security:** Adheres to Salesforce's comprehensive security model, ensuring data integrity.
4. **Scalable Solutions:** Capable of handling both simple and enterprise-grade solutions.
5. **Testing Requirements:** Built-in testing framework ensures code reliability before deployment.

Limitations

1. **Governor Limits:**
    - Ensures resources are shared fairly in Salesforce’s multi-tenant environment, but it can limit execution if not managed well.
2. **Platform-Dependency:**
    - Apex can only run on the Salesforce platform, which may not suit all use cases.
3. **Learning Curve:**
    - Developers unfamiliar with Salesforce-specific paradigms like governor limits or data structures may face initial challenges.

Getting Started with Apex

1. **Prerequisites:**
   - Familiarity with Salesforce platform basics and object-oriented programming concepts.
2. **Setup:**
   - Use Salesforce's Developer Console, Visual Studio Code (with Salesforce Extensions), or other IDEs.
   - Create a Salesforce Developer Edition account to access a free playground for experimentation.

💡 Theory for Viva – Apex, Salesforce, and Cloud

🔹 1. What is Salesforce?

Answer:

Salesforce is a cloud-based CRM (Customer Relationship Management) platform that allows businesses to manage customer data, automate processes, and build custom applications using low-code and code-based tools.

🔹 2. What is Apex in Salesforce?

Answer:

Apex is a strongly-typed, object-oriented programming language used in Salesforce.

It is syntactically similar to Java.

Apex is used for writing business logic, triggers, web services, scheduled jobs, and custom controllers.

🔹 3. Where is Apex code executed?

Answer:

Apex code is executed on Salesforce's multi-tenant cloud platform in a hosted environment called the Lightning Platform / Force.com.

🔹 4. How do you run Apex code?

Answer:

Apex code can be run in multiple ways:

Execute Anonymous Window (Developer Console) – for testing code

Triggers – for automating data operations

Classes and Methods – for reusable logic

Visualforce or Lightning Components – for UI-based apps

🔹 5. What is the Developer Console in Salesforce?

Answer:

The Developer Console is a browser-based tool in Salesforce used for:

Writing and running Apex code

Debugging with logs

Running SOQL queries

Managing tests

🔹 6. What are Static Methods in Apex?

Answer:

A static method:

Belongs to the class, not objects

Can be called directly like ClassName.methodName()

Is used for utility or helper functions

- 7. What is System.debug()?

Answer:

System.debug() is used to:

Print output to the Debug Logs

Help in troubleshooting and testing

It doesn’t display to users, only to developers

- 8. What are some benefits of using Apex?

Tight integration with Salesforce data

Built-in security and governor limits

Can be used with Lightning and Visualforce

Allows automation of complex business logic

- 9. What are Governor Limits in Apex?

Answer:

Since Apex runs in a shared cloud environment, Salesforce enforces limits on:

CPU time

SOQL queries

Number of records processed

To ensure fair use and avoid monopolizing server resources.

- 10. What is Force.com Platform (Lightning Platform)?

Answer:

It’s the PaaS (Platform-as-a-Service) layer of Salesforce that lets you:

Build custom applications

Write business logic using Apex

Create custom UIs with Visualforce or Lightning Components